



Evaluating expressions without changing the subject

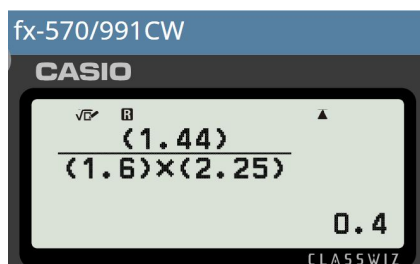
Task A: Rearranging in order to solve

1) The formula for volume of a cuboid can be written as $V = lwh$

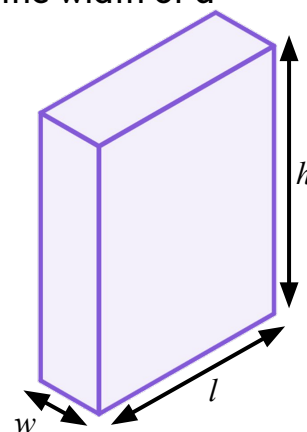
Rearrange the formula to make w the subject. Then evaluate to find the width of a cuboid with length 1.6 cm, height 2.25 cm, and volume 1.44 cm³

$$V = lwh$$

$$\frac{V}{lh} = w$$



$$w = 0.4 \text{ cm}$$



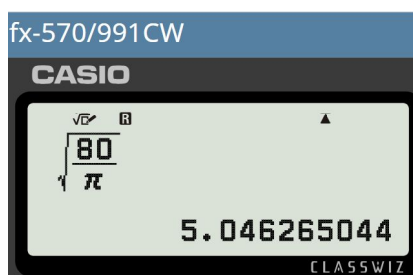
2) The formula for the area of a circle is $A = \pi r^2$

Rearrange the formula to make r the subject. Then evaluate to find the radius of a circle with area 80 cm²

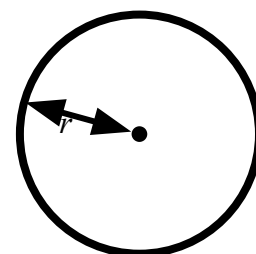
$$A = \pi r^2$$

$$\frac{A}{\pi} = r^2$$

$$\sqrt{\frac{A}{\pi}} = r$$



$$r = 5.05 \text{ cm (to 2 d.p.)}$$



Task B: Evaluating without changing the subject

1) Missing dimensions in a trapezium.

a) A trapezium has area 95 cm², a lower base length of 23 cm and an upper base length of 15 cm

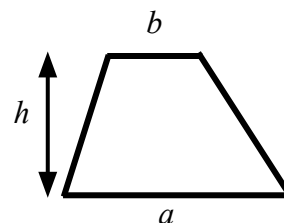
Substitute without rearranging to find the height (h).

$$A = \frac{1}{2}(a + b)h$$

$$95 = \frac{1}{2}(23 + 15)h$$

$$95 = 19h$$

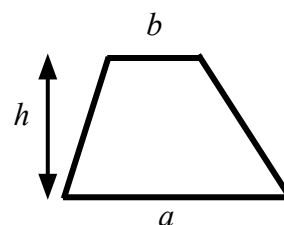
$$h = 5 \text{ cm}$$



$$A = \frac{1}{2}(a + b)h$$

b) A trapezium has area 88 cm^2 , a lower base length of 14 cm and a height of 8 cm

Substitute without rearranging to find the upper base (b).



$$A = \frac{1}{2}(a + b)h$$

$$88 = \frac{1}{2}(14 + b)8$$

$$88 = 4(14 + b)$$

$$88 = 56 + 4b$$

$$32 = 4b$$

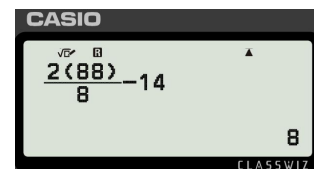
$$b = 8 \text{ cm}$$

An alternative method would be to rearrange the formula to make b the subject...

$$A = \frac{1}{2}(a + b)h$$

$$\frac{2A}{h} - a = b$$

...and substitute.

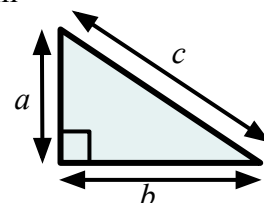


c) Compare the approaches in parts a) and b)

Part (b) was the harder problem. It had more steps and involved more skills, therefore more opportunities to make an error.

2) A right triangle has hypotenuse 13 cm , and one shorter length of 12 cm

a) Substitute and evaluate without rearranging to find the remaining shorter length.



$$a^2 + 12^2 = 13^2$$

$$a^2 = 13^2 - 12^2$$

$$a^2 = 169 - 144$$

$$a^2 = 25$$

$$a = 5 \text{ cm}$$

$$a^2 + b^2 = c^2$$

b) Rearrange to make a the subject, then substitute to find the remaining shorter length.

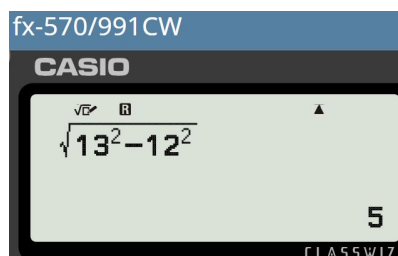
$$a^2 + b^2 = c^2$$

$$a^2 = c^2 - b^2$$

$$a = \sqrt{c^2 - b^2}$$

$$a = \sqrt{13^2 - 12^2}$$

$$a = 5 \text{ cm}$$



c) Compare the approaches in parts a) and b)

There's a lot of similarity between the two methods. They give us the same result with a similar level of effort in this problem.

